

Requirements Document -- AI Inventory Agent

1. Functional Requirements

1.1 Authentication & Authorization

ID	Requirement
FR-AUTH-01	The system shall allow users to register with user
FR-AUTH-02	The system shall authenticate users via username a
FR-AUTH-03	The system shall hash passwords using BCrypt befor
FR-AUTH-04	JWT tokens shall expire after 24 hours
FR-AUTH-05	The system shall reject requests with invalid or e
FR-AUTH-06	The frontend shall store auth tokens in sessionSto
FR-AUTH-07	The system shall prevent duplicate usernames and e

1.2 Product Management

ID	Requirement
FR-PROD-01	The system shall support full CRUD operations for
FR-PROD-02	Each product shall have: name (required, max 100 c
FR-PROD-03	The system shall auto-generate timestamps (created
FR-PROD-04	The system shall support product search by name or
FR-PROD-05	The system shall provide a low-stock endpoint retu
FR-PROD-06	The system shall allow toggling a product's active
FR-PROD-07	The frontend shall display products in a table wit
FR-PROD-08	The system shall validate all inputs using Jakarta

1.3 Inventory Management

ID	Requirement
FR-INV-01	The system shall record inventory movements with t
FR-INV-02	The system shall calculate current stock per produ
FR-INV-03	The frontend shall display stock levels with movem
FR-INV-04	The system shall filter movements by type, product

1.4 Supplier Management

ID	Requirement
FR-SUP-01	The system shall support full CRUD operations for
FR-SUP-02	Each supplier shall have: name (required, max 100
FR-SUP-03	The frontend shall support filtering suppliers by

1.5 Sales Management

ID	Requirement
FR-SALE-01	The system shall record sales with product, quanti
FR-SALE-02	The system shall auto-calculate totalPrice (unitPr
FR-SALE-03	Sale status shall default to PENDING and support C
FR-SALE-04	The frontend shall filter sales by customer, produ

1.6 Purchase Management

ID	Requirement
FR-PUR-01	The system shall create purchase orders linked to
FR-PUR-02	Purchase status shall default to ORDERED and suppo
FR-PUR-03	The system shall auto-calculate totalCost (unitCos
FR-PUR-04	The frontend shall allow receiving (ORDERED -> REC
FR-PUR-05	The frontend shall filter purchases by supplier, s

1.7 Dashboard

ID	Requirement
FR-DASH-01	The dashboard shall display KPI cards: total produ
FR-DASH-02	The dashboard shall show a sales trend line chart
FR-DASH-03	The dashboard shall show a category distribution p
FR-DASH-04	The dashboard shall show a top products bar chart
FR-DASH-05	The dashboard shall display a recent activity feed

1.8 AI Features

ID	Requirement
FR-AI-01	The system shall forecast 7-day demand per product
FR-AI-02	The system shall compute a stock health score (0-1
FR-AI-03	The system shall detect anomalies in sales quantit
FR-AI-04	The system shall generate reorder suggestions with
FR-AI-05	The system shall accept natural language queries a

1.9 Reports

ID	Requirement
FR-RPT-01	The reports page shall display aggregated sales, p
FR-RPT-02	The reports page shall render Chart.js visualizati

2. Non-Functional Requirements

2.1 Performance

ID	Requirement
NFR-PERF-01	API responses shall return within 2 seconds under
NFR-PERF-02	The MySQL connection pool shall support up to 20 c
NFR-PERF-03	AI service responses shall return within 5 seconds

2.2 Security

ID	Requirement
NFR-SEC-01	All API endpoints except auth shall require a vali
NFR-SEC-02	Passwords shall be stored as BCrypt hashes (never
NFR-SEC-03	CORS shall allow all origins for development; rest
NFR-SEC-04	The JWT secret key shall be configurable via appli
NFR-SEC-05	Sensitive configuration (DB credentials, JWT secre

2.3 Reliability

ID	Requirement
NFR-REL-01	The system shall handle entity-not-found errors gr

NFR-REL-02	The system shall validate all request inputs and r
NFR-REL-03	The global exception handler shall return structur

2.4 Usability

ID	Requirement
NFR-USE-01	The frontend shall be responsive and work on desk
NFR-USE-02	All data pages shall include collapsible filter pa
NFR-USE-03	The system shall display toast notifications for s
NFR-USE-04	The sidebar shall highlight the current active pag

2.5 Maintainability

ID	Requirement
NFR-MAINT-01	The codebase shall follow a layered architecture:
NFR-MAINT-02	DTOs shall separate API contracts from entity inte
NFR-MAINT-03	Each AI module shall be an independent Python serv
NFR-MAINT-04	Frontend JavaScript shall be organized per-page wi

2.6 Compatibility

ID	Requirement
NFR-COMPAT-01	The backend shall run on Java 21
NFR-COMPAT-02	The AI service shall run on Python 3.14+
NFR-COMPAT-03	The frontend shall work in Chrome, Firefox, and Ed
NFR-COMPAT-04	The database shall be MySQL 8.0 via XAMPP

3. System Constraints

- Single-user scope: The system is designed for a single organization; no multi-tenancy
- No payment integration: Sales and purchases are recorded manually
- AI service is optional: The frontend and backend work independently of the Flask AI service
- Local development only: No containerization, CI/CD, or cloud deployment configured